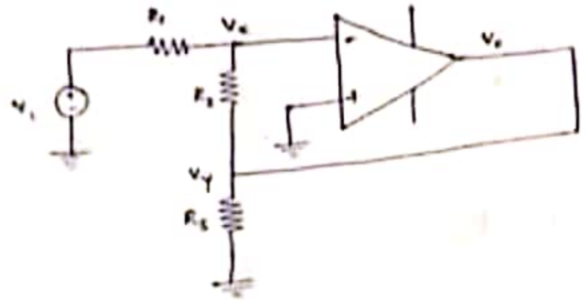
ABHINAV RAJ ELECTRONICS ASSIGNMENT 18JE0018

1.



$V_x = 0$

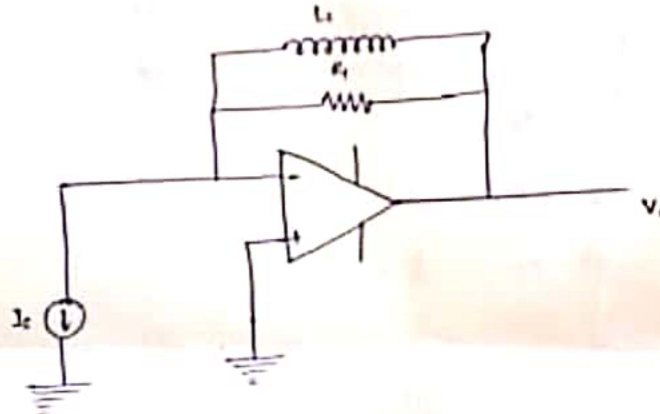
$V_y = V_o$

Nodal at x : $\dfrac{V_x - V_1}{R_1} + \dfrac{V_x - V_o}{R_2} = 0$

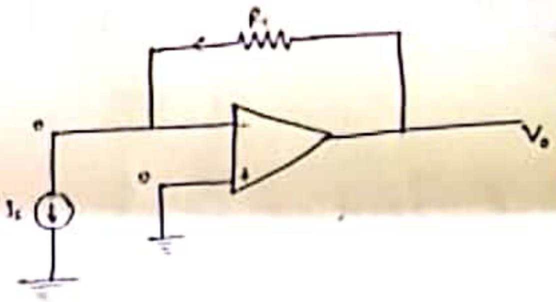
$\Rightarrow -\dfrac{V_1}{R_1} - \dfrac{V_o}{R_2} = 0 \Rightarrow V_o = -\dfrac{R_2}{R_1} V_1$

Voltage gain $= \dfrac{-R_2}{R_1}$

2.

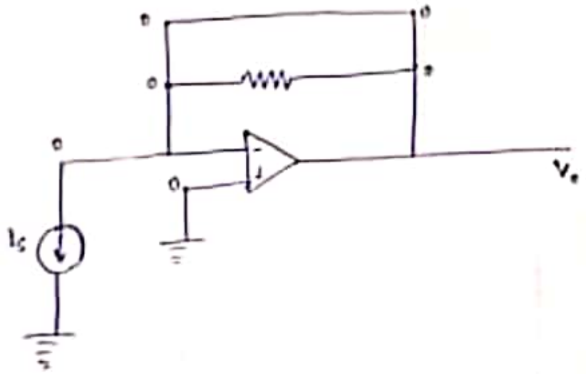


at low frequency:



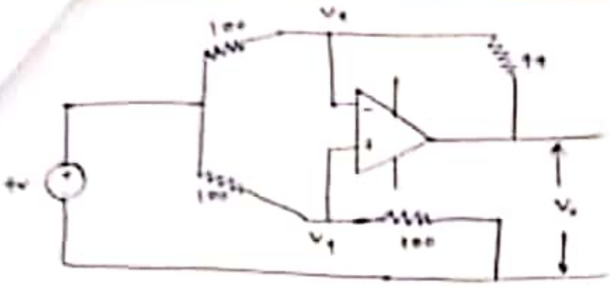
$V_o = I_s R_1$

at high frequency:



$V_o = 0V$

Hence, low pass filter.



$V_x = V_y$

at node x :

$$\frac{4 - V_x}{100} = \frac{V_x - V_o}{99}$$

at node y

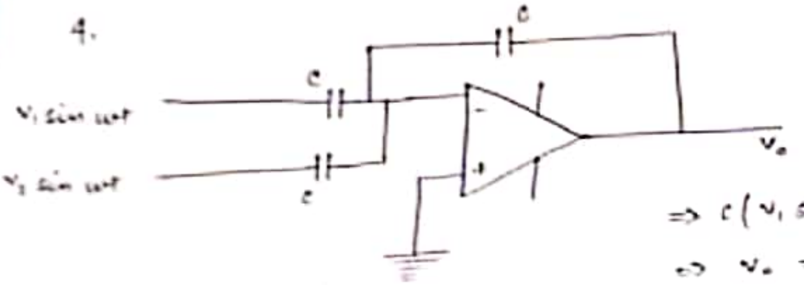
$$\frac{4 - V_y}{100} = \frac{V_x}{100}$$

$V_x = V_y$: $\frac{4 - V_y}{100} = \frac{V_y}{100} \Rightarrow V_y = 2V$

$$\frac{4-2}{100} = \frac{2 - V_o}{99} \Rightarrow V_o = 2 - 99(0.02)$$

$= 0.02$ V (Ans)

4.

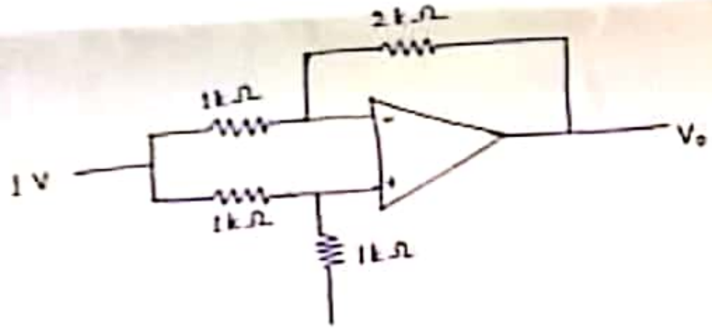


charge of capacitor 1 + charge of capacitor 2 = charge of capacitor 3

$\Rightarrow c(v_1 \sin wt) + c(v_2 \sin wt) = c(0 - v_o)$

$\Rightarrow v_o = -(v_1 \sin wt + v_2 \sin wt)$

5.

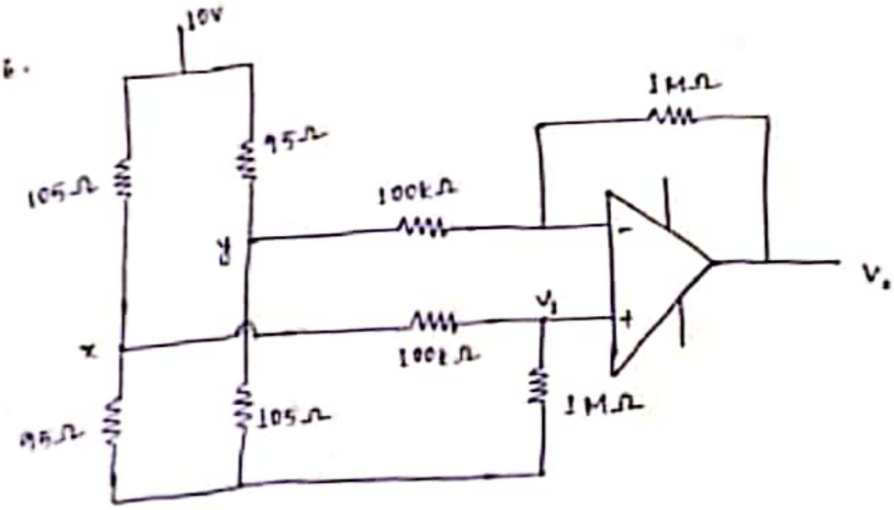


$V_o = -\frac{2}{1}(1) + \left(1 + \frac{2}{1}\right) \times V_1$

$\Rightarrow V_o = -2 + 3 \times 1 \times \frac{1}{2}$

$= -2 + 1.5 \Rightarrow \boxed{V_o = -0.5\,V}$

6.



$$V_1 = \frac{x \times 10^3}{10^3 + 100} = 0.9x$$

$\Rightarrow x = 1.1 V_1$

Applying nodal at x

$$\frac{x-10}{105} + \frac{x}{95} + \frac{x-V_1}{100} = 0$$

$$\Rightarrow \frac{1.1V_1 - 10}{105} + \frac{1.1V_1}{95} + \frac{1.1V_1 - V_1}{100} = 0$$

$$\Rightarrow V_1\left[\frac{1.1}{105} + \frac{1.1}{95} + \frac{0.1}{100}\right] = \frac{10}{105} \Rightarrow \boxed{V_1 = 4.13V}$$

Applying nodal at y:

$$\frac{y-10}{95} + \frac{y}{105} + \frac{y-V_1}{100} = 0$$

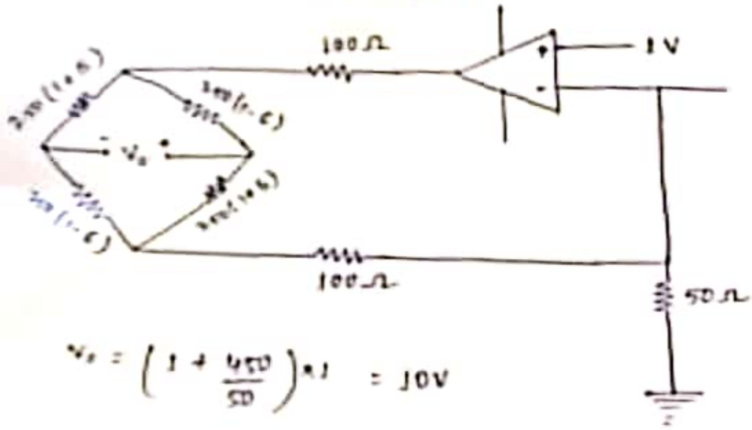
$$\Rightarrow \frac{y-10}{95} + \frac{y}{105} + \frac{y-4.13}{100} = 0$$

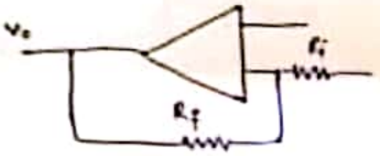
$$\Rightarrow y = \frac{\frac{10}{95} + \frac{4.13}{100}}{\left(\frac{1}{95} + \frac{1}{105} + \frac{1}{100}\right)} = \boxed{4.87V}$$

$$V_0 = -\frac{10^3}{100} \times 4.87 + \left(1 + \frac{10^3}{100}\right) \times 4.13$$

$V_0 = -3.27V$ (Ans)

7.



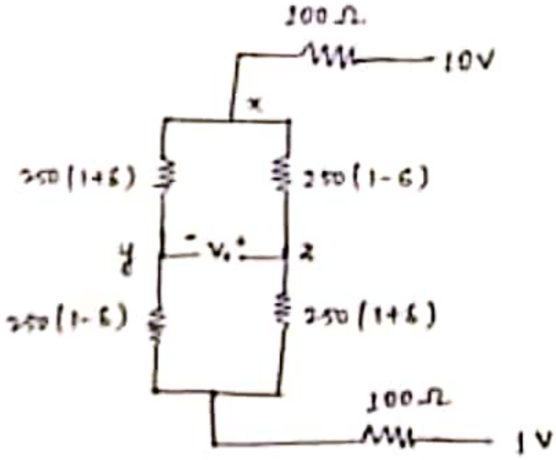


$$V_0 = \left(1 + \frac{450}{50}\right) \times 1 = 10V$$

$R_i = 50\Omega$

$R_f = 450\Omega$

$\delta = 0.05V$ given

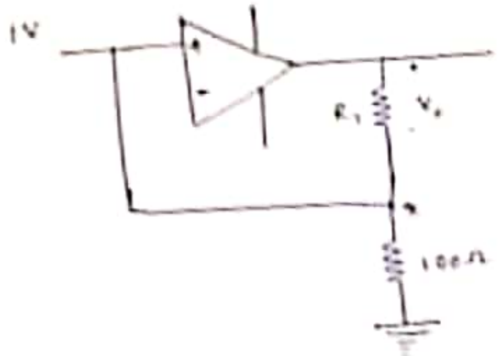


$$i = \frac{9}{450} = 0.02 A$$

$$V_y = 10 - 100 \times 0.02 - 0.01 \times 250(1+\delta)$$
$$= 5.5 - 0.01\delta$$

$$V_z = 10 - 100 \times 0.02 - 0.01 \times 250(1-\delta)$$
$$= 5.5 + 0.01\delta$$

$$\Rightarrow V_0 = V_z - V_y$$
$$= 0.02\delta$$
$$= 0.02 \times 0.05$$
$$= 0.0010V$$



since op-amp is ideal

$\boxed{V_o = 1V}$

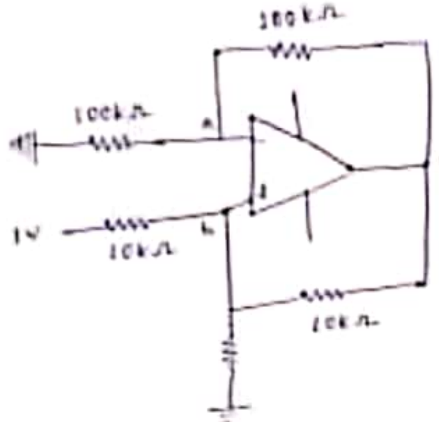
applying KVL in loop (1)

$-V_o - V_R = 0$

$\Rightarrow V_o = -V_R$

$\Rightarrow \boxed{V_o = -1V}$

9.



Applying nodal at b:

$\frac{V_x - 1}{10} + \frac{V_x}{R_L} + \frac{V_x - V_o}{10} = 0$

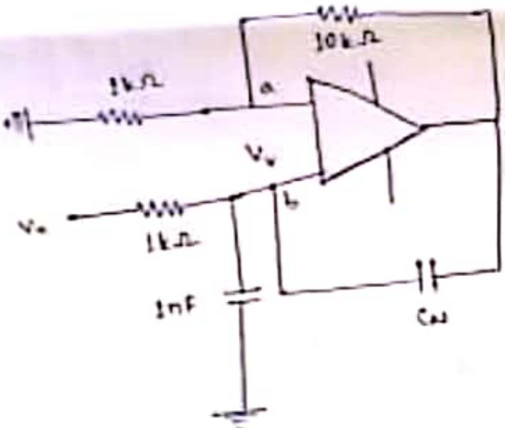
$\Rightarrow \left(\frac{V_x}{10} + \frac{V_x - V_o}{10}\right) - \frac{1}{10} + \frac{V_x}{R_L} = 0$

$\Rightarrow 0 - \frac{1}{10} + \frac{V_x}{R_L} = 0$

$\boxed{\frac{V_x}{R_L} = I_L} \Rightarrow I_L = 0.1mA \Rightarrow \boxed{I_L = 100\mu A}$

Applying nodal at a: $\frac{V_x}{100} + \frac{V_x - V_o}{100} = 0 \Rightarrow \frac{V_x}{10} + \frac{V_x - V_o}{10} = 0$

10.



Applying nodal at A:

$\frac{V_v}{1} + \frac{V_v - V_o}{10} = 0$

$\Rightarrow V_o = (1+10)V_v = 11V_v$

Applying Nodal at b:

$(V_x - V_v)\times 10^{-3} = 10^{-9}\frac{d}{dt}(V_v) + \frac{d}{dt}(V_v - V_o)C_N$

$= 10^{-9}\frac{d}{dt}V_v - \frac{d}{dt}10V_vC_N$

$(V_x - V_v)\times 10^{-3} = (10^{-9} - 10C_N)\frac{d}{dt}V_v$

Given $V_x = V_v$

$10^{-9} - 10C_N = 0$

$C_N = 10^{-10}$ F

$\because 1pF = 10^{-12}$ F

$\boxed{C_N = 100pF}$